

A PROJECT REPORT ON

**Covid-19 Protocol Detection System Using
Deep Learning**

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**BACHELOR OF ENGINEERING
(Computer Engineering)**

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Abstract

The Covid-19 had a huge impact on different sectors in many countries and such impact caused problems to many people around the world. Alleviation in such pandemic can be solved by social distancing as well as putting on its face mask. This small step of wearing a face mask as well as following social distancing would save lots of lives as the spread of the virus could be mitigated. This project is implemented by considering these facts of Covid-19 protocols. The project aims to detect face mask, social distancing on video feed and also reminder for vaccination by using deep learning models.

SSD algorithm is used for face mask detection and if mask is not wear then recognition of face will be done by using face recognition algorithm. YOLO object detection is used to detect the people in frame and distance is measured by using Euclidean formula. This project also has one more feature that is detection for vaccination, so next dose will be calculated and reminder will be sent to user. Also developed a desktop application where admin will get notified and can monitor the violations. If violation happens, data will get stored into firebase database and it will also notify admin. The desktop application has features of viewing history of violated data, notification functionality, user registration, etc. The overall accuracy of our system is above 90% depending on the quality of video we give as a input.

Keywords

YOLO Object Detection, Firebase, SSD, face recognition

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CHAPTER 1

INTRODUCTION

1.1 Motivation

In world, one of the major concern is spreading covid-19 so fast and not following the rules leads to increase in cases. The biggest challenge for private sector like company, colleges is to follow the rules and employees or students can do their work. So we are decided to develop a system which detects the violation for covid-19 protocol. So it will help people to follow the rules and also for the organization to discipline the rules.

1.2 Problem Definition and Objectives

1.2.1 Problem Definition

To implementing Covid-19 system using deep learning for identifying face mask, social distancing and vaccination along with detection and notifying functionality.

1.2.2 Objectives

- To detect whether person has wear face mask properly or not.
- To reduce violation of Social Distancing rules.
- To identify person is vaccinated or not.
- Notifying organization.

1.3 Project Scope

This application specially design for private sectors like

- Offices
- Companies
- College Campus
- Apartments Etc.

CHAPTER 2

LITERATURE SURVEY

No	Title	Author,Publisher and Year	Technique	Remark
1	Covid-19 Face Mask Detection Using TensorFlow, Keras and OpenCV	Arjya Das, Mohammad Wasif Ansari and Rohini Basak, 2020	Keras and OpenCV	Keras and OpenCV are used with the accuracy of 94%
2	Social Distancing Detection with Deep Learning Model	Yew Cheong Hou, Mohd Zafri Baharuddin, Salman Yussof, Sumayyah Dzulkifly2020	YOLO v3	YOLO v3 is used with the detection accuracy of 90%
3	Real-Time Implementation Of Face Recognition System	Neel Ramakant Borkar, Sonia Kuwelkar, 2017	PCA, LDA	PCA and LDA algorithm are used with the accuracy of 91%

Table 2.1: Literature survey

2.1 Covid-19 Face Mask Detection Using TensorFlow, Keras and OpenCV

Author : Arjya Das, Mohammad Wasif Ansari and Rohini Basak

Summary : COVID-19 pandemic has rapidly affected our day-to-day life disrupting the world trade and movements. Wearing a protective face mask has become a new normal. In the near future, many public service providers will ask the customers to wear masks correctly to avail of their services. Therefore, face mask detection has become a crucial task to help global society. This paper presents a simplified approach to achieve this purpose using some basic Machine Learning packages like TensorFlow, Keras, OpenCV and Scikit-Learn. The proposed method detects the face from the image correctly and then identifies if it has a mask on it or not. As a surveillance task performer, it can also detect a face along with a mask in motion.

2.2 Social Distancing Detection with Deep Learning Model

Author : Yew Cheong Hou¹, Mohd Zafri Baharuddin, Salman Yussof, Sumayyah Dzulkifly

Summary : The paper presents a methodology for social distancing detection using deep learning to evaluate the distance between people to mitigate the impact of this coronavirus pandemic. The detection tool was developed to alert people to maintain a safe distance with each other by evaluating a video feed. The video frame from the camera was used as input, and the open-source object detection pre-trained model based on the YOLOv3 algorithm was employed for pedestrian detection. Later, the video frame was transformed into top-down view for distance measurement from the 2D plane. The distance between people can be estimated and any non-compliant pair of people in the display will be indicated with a red frame and red line.

2.3 Real-Time Implementation Of Face Recognition System

Author : Neel Ramakant Borkar, Sonia Kuwelkar

Summary : Face Recognition is the ability to detect and recognize a person by their facial characteristics. Face is a multidimensional and hence requires

a lot of mathematical computations. Face recognition system is very essential and important for providing security, mug shot matching, law enforcement applications, user verification, user access control, etc and is mostly used for recognition for various applications. These all applications require an efficient Face recognition system. There are many methods that are already proposed and have low recognition capability, high false alarm rate. Hence the major task of the research is to develop face recognition system with improved accuracy and improved recognition time of an face recognition system. This paper proposes a hybrid face recognition algorithm by combining two face recognition techniques by integrating (PCA) principle Component Analysis, (LDA) Linear Discriminant Analysis.

CHAPTER 3
SOFTWARE REQUIREMENTS
SPECIFICATION

3.1 Assumptions and Dependencies

For developing this system we are assuming that system has the information about people in organization. And also there is a person from organization as an admin who will constantly monitor this through our system.

3.2 Functional Requirements

3.2.1 Face Mask Detection

For detection of face mask required modules like 'Face Recognition' and 'SSD' are loaded into the system. 'SSD Module' plays an important role to check whether mask is wear properly or not. If not, the module will detect user and notify to that user. SSD stands for 'Single Shot Multibox Detector'. It is a technique that is used to detect objects in images using a single deep neural network. Basically it is used for object detection in an image. We are using SSD for detecting mask as an object in our system. For this purpose we are labelling our image dataset (with-mask and without-mask). By labelling it means we are plotting co-ordinates for the position of mask over face. As these co-ordinates in turn will be used to detect whether person has wear mask properly or not.

3.2.2 Face Recognition

For detection of user, firstly User dataset will be loaded which is needed to train the model. Then we take the frame as input from the face mask detection module. After getting the frame, detection of face is done using SSD, extracting features of detected face is done using Face recognition. Finally classifying the extracted feature and then getting the details of the person who has not wear the mask properly. Once we get the details of that person we will notify that user. There is a possibility that the user may avoid our notification, so we calculate how often user is breaking the rules. Once the count for breaking the rule goes beyond standard count, system will notify to the organization.

3.2.3 Social Distancing

Social Distancing is methodology in which humans stay away from each other physically. For example, in current COVID-19 situation it is being followed widely, in order to prevent the spread of the disease. Our system detects

whether people follows social distancing rule or not and also notify organization using deep learning. In this, firstly people in frame are detected using YOLO object detection algorithm. After detection of people, centroid point is calculated based on the Boundary Box. Then the distance between two centroid (people) calculated by using Euclidean distance formula. That distance is compare with standard distance and if it is less then it will create a Red Boundary Box around them. If the total number of red Boundary boxes is maximum than standard count then social distancing rule is violated and it will notify the organization.

3.2.4 Vaccination

In vaccination module, it will use the User dataset. Based on the previous date of dose system will calculate next date of dose for every record in dataset. After that system will remind the user some days before the actual date of next dose.

3.2.5 Alert Messages

In this feature, alert message will get send via e-mail to person in organization who violates Covid-19 rules. Also it will be displayed on the admin panel so that admin can come to know who violated the rule and keep the record.

3.3 External Interface Requirements

3.3.1 User Interfaces

We have developed UI so that Admin can easily monitor people through it. The interface shows information such as violated data, notification regarding rule violation and also have functionality for user registration.

3.3.2 Hardware Interfaces

3.3.2.1 External Camera

For capturing live data, system also provides capability to connect external camera. So that system can get live data and it will start to work in real-time

3.4 Nonfunctional Requirements

3.4.1 Performance Requirements

For achieving better performance, quality of input video should be high. In case of external camera the capturing quality should be high.

3.4.1.1 Protection for system generated data

Frames will get stored in the form of zip file which are password protected and this password is only known to the Admin.

3.4.1.2 Access Control

Only the Admin has full access over the folder where violated frames are stored. Only Admin has full access over system.

3.4.2 Software Quality Attributes

- Testability
- Reliability
- Usability
- Extensibility

3.5 System Requirements

3.5.1 Database Requirements

3.5.1.1 Firebase

- Real-time Database:

Light weight data like text data will get stored.

- Firebase Storage:

Data having large size will get stored.

3.5.2 Software Requirements

- Python version 3.8 and above
- Microsoft Visual Studio 2022
- Pycharm 2020.1.3 Community Edition

3.5.3 Hardware Requirements

- PC with min I3 Processor
- 8 GB RAM
- 500 GB HDD
- At least 2GB Of VRAM

3.6 Analysis Models:SDLC Model to be applied

Incremental Model

It generates working software quickly and early during the software life cycle. Incremental Model is a process of software development where requirements are broken down into multiple standalone modules of software development cycle. Incremental development is done in steps from analysis design, implementation, testing/verification, maintenance.

Each iteration passes through the requirements, design, coding and testing phases. And each subsequent release of the system adds function to the previous release until all designed functionality has been implemented. The system is put into production when the first increment is delivered. The first increment is often a core product where the basic requirements are addressed, and supplementary features are added in the next increments. Once the core product is analysed by the client, there is plan development for the next increment.

Characteristics of an Incremental module includes:

- System development is broken down into many mini development projects.
- Partial systems are successively built to produce a final total system.
- Highest priority requirement is tackled first.
- Once the requirement is developed, requirement for that increment are frozen.

Incremental Phases includes:

- Requirement Analysis: Requirement and specification of the software are collected.
- Design : Some high-end function are designed during this stage.
- Code : Coding of software is done during this stage.
- Test : Once the system is deployed, it goes through the testing phase.

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

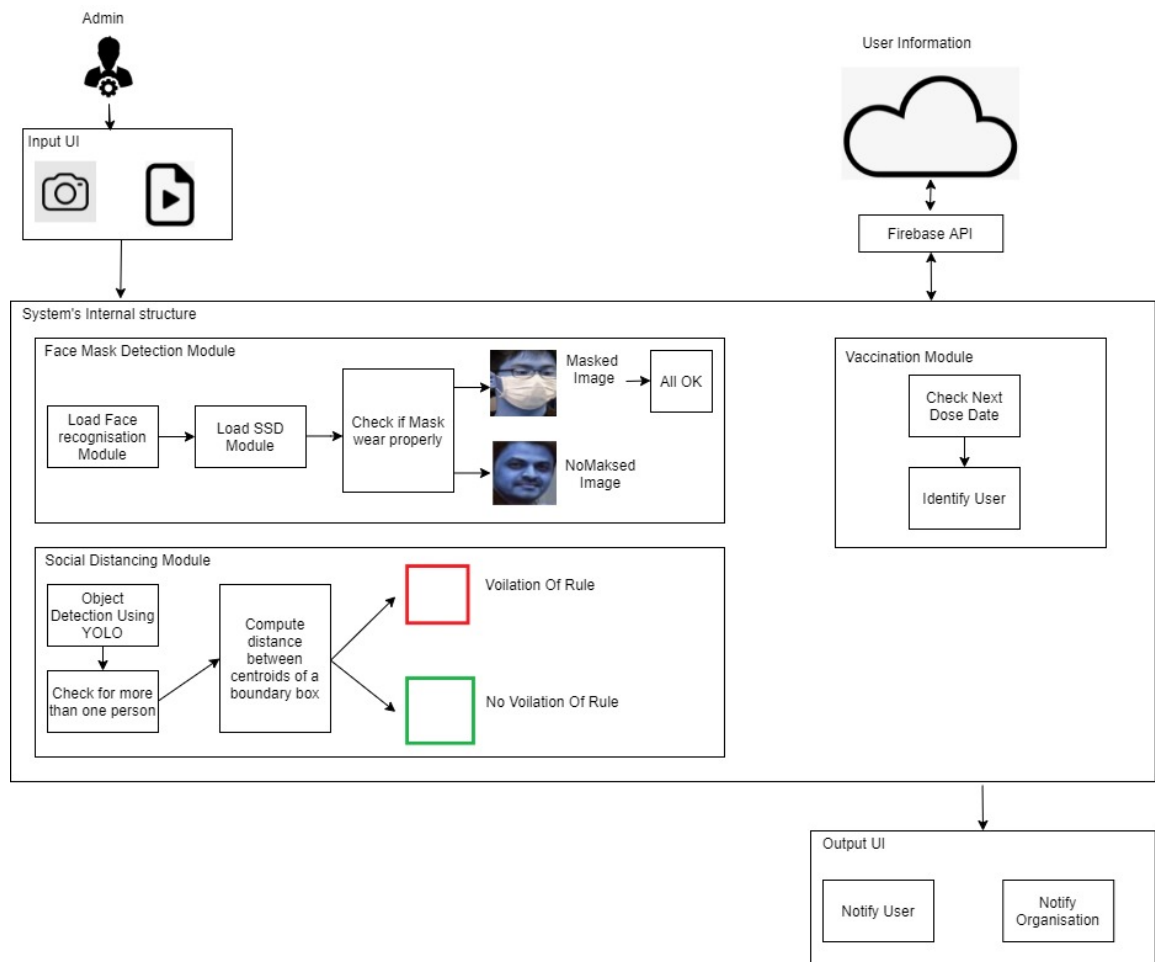


Figure 4.1: Architecture Diagram

4.2 Mathematical Model

4.2.1 Centroid calculation and Euclidean distance formula

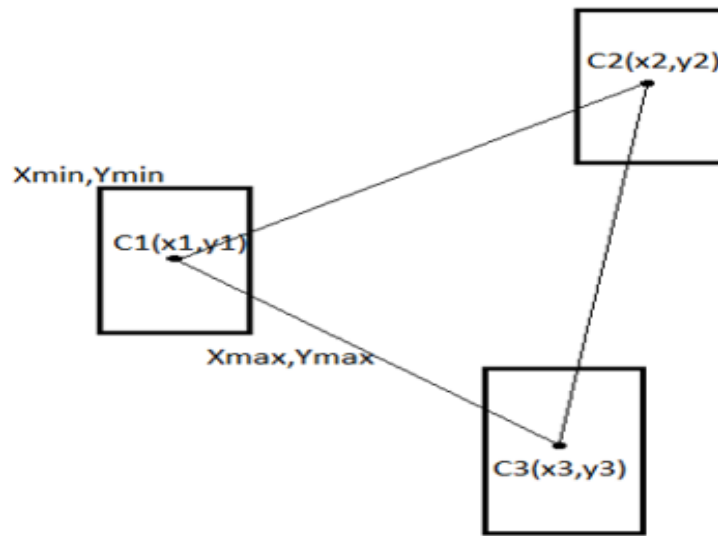


Figure 4.2: Mathematical Model-1

In Social distancing module, after detecting people from frames and drawing rectangular box for them. Next step is to calculate the centroid for each person so that it will be used in distance calculation.

Formula for calculate Centroid :

$$C1(x1, y1) = (Xmin + Xmax)/2, (Ymin + Ymax)/2$$

After that we have calculate the distance between two centroid using euclidean distance

Formula for calculating Euclidean Distance :

$$\sqrt{(x2 - x1)^2 + (y2 - y1)^2}$$

4.2.2 Non Max Supression

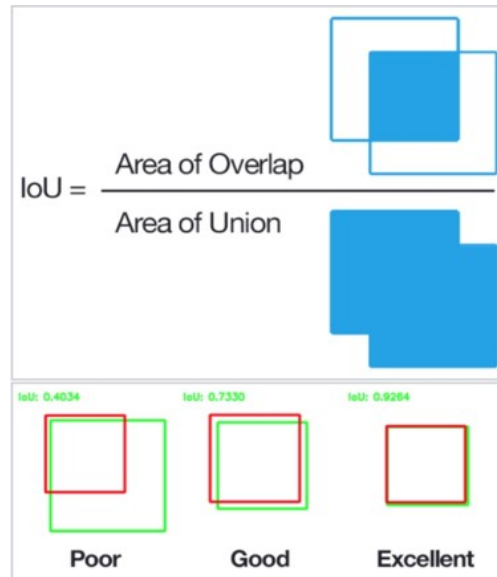


Figure 4.3: Mathematical Model-2

The purpose of non-max suppression is to select the best bounding box for an object and reject or “suppress” all other bounding boxes. The NMS takes two things into account

- The objectiveness score is given by the model
- The overlap or IOU of the bounding boxes

Intersection Over Union (IoU) :

The Intersection over Union (IoU) metric is essentially a method used usually to quantify the percent overlap between the ground truth BBox (Bounding Box) and the prediction BBox. However, in NMS, we find IoU between two predictions BBoxes instead.

4.3 Data Flow Diagrams / UML Diagrams

4.3.1 Data Flow Diagram

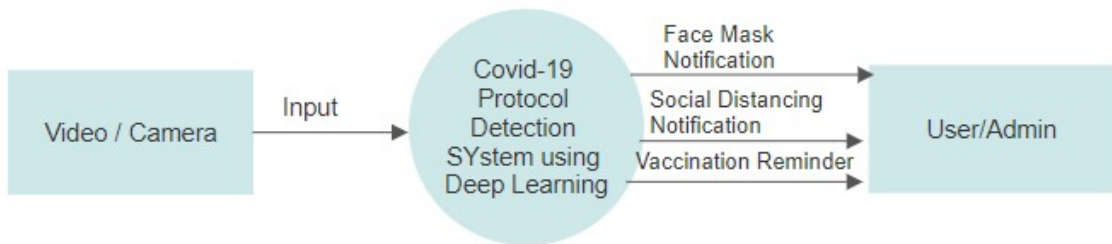


Figure 4.4: DFD Level 0

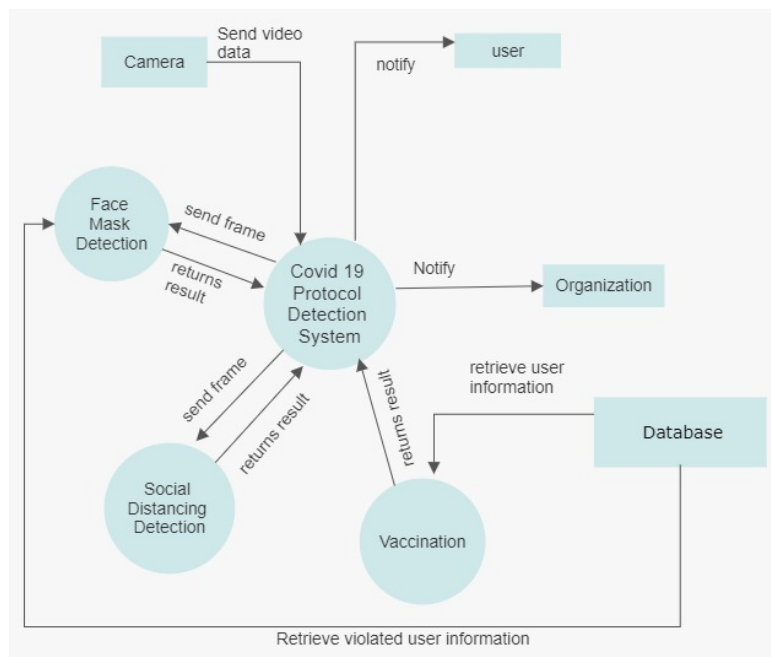


Figure 4.5: DFD Level 1

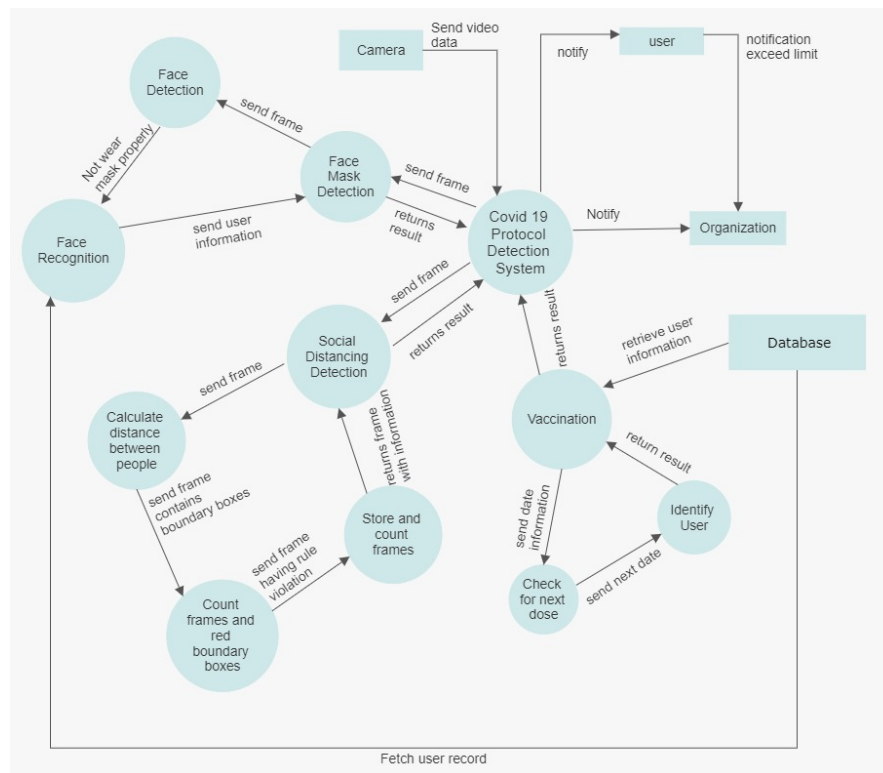
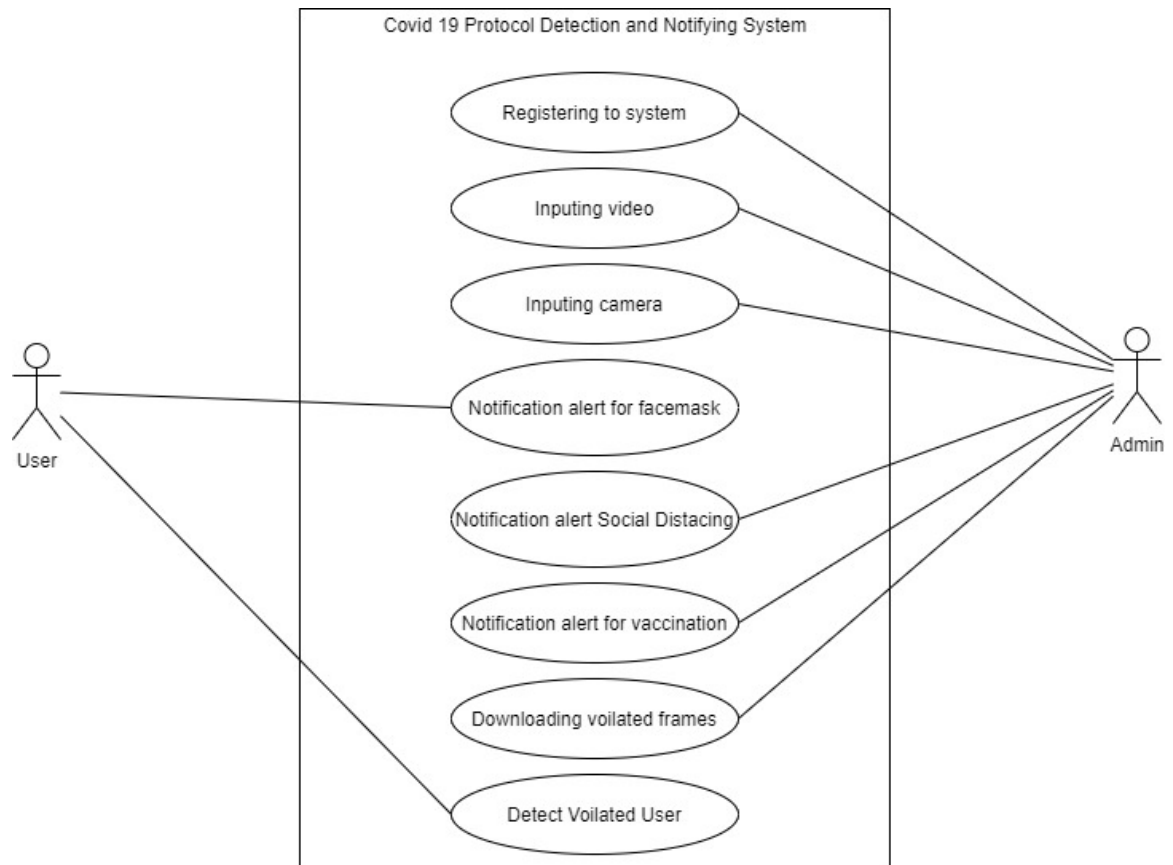


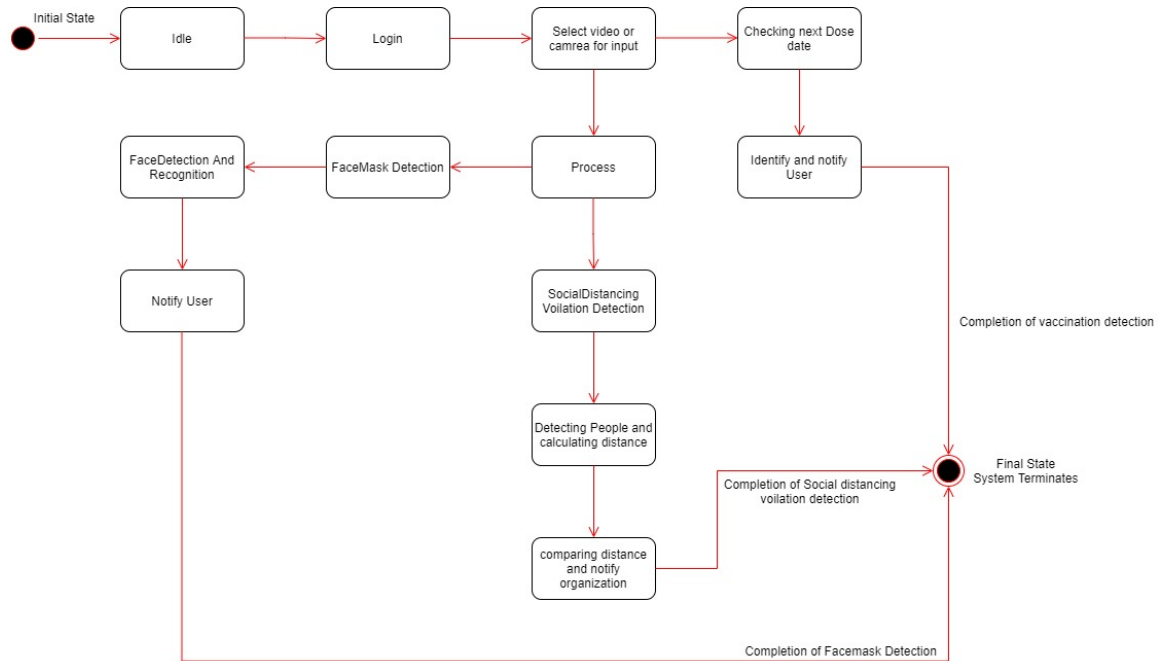
Figure 4.6: DFD Level 2

4.3.2 UML Diagrams

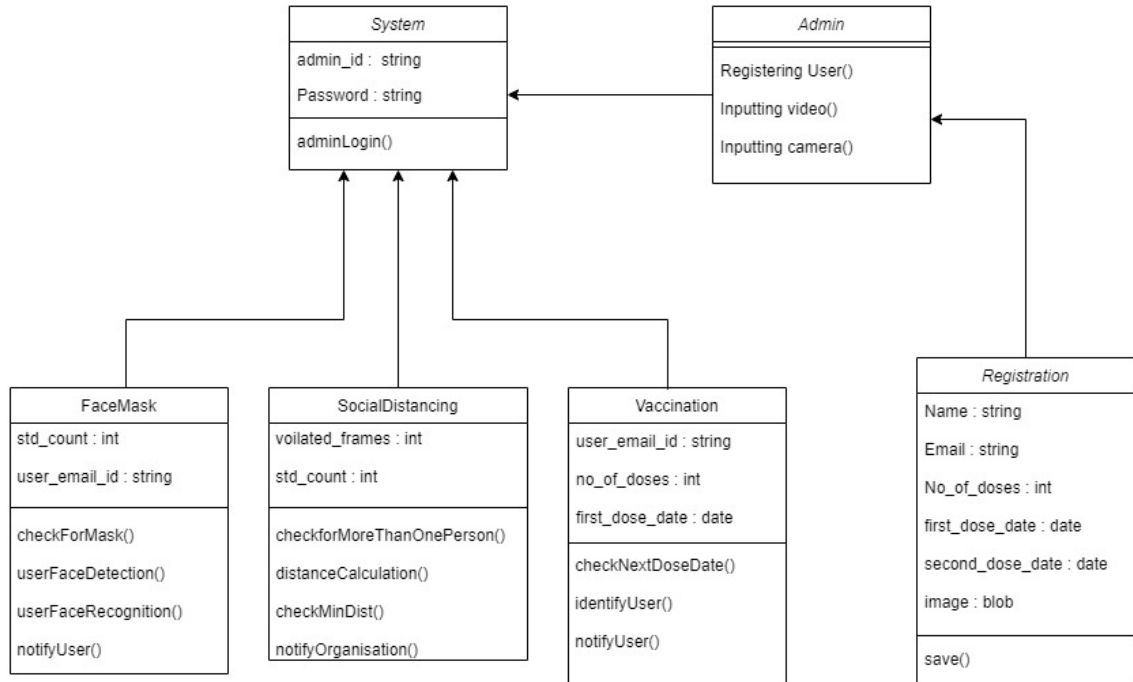
4.3.2.1 Use Case Diagram



4.3.2.2 State Diagram



4.3.2.3 Class Diagram



CHAPTER 5

PROJECT PLAN

5.1 Project Estimate

5.1.1 Reconciled Estimates

- Cost Estimate: The software used for the application will incur no cost as it is all released under free open source licenses.
- Time Estimate: 9-10 months

5.1.2 Human Resources

- Number of people : 4
- Skills: Python, Js, Firebase APIs
- Geographic restrictions: None
- Client: In our project, the clients are private companies, apartments, institutes.
- Stakeholders: Stakeholders are often members of the project team. Considering the project, stakeholders will be the people who will be using the application, i.e, the clients along with project team.

5.1.3 Development Resources

- Data Collection: Through user registrations
- Software:
 1. Python libraries
 2. PyCharm IDEA
 3. Firebase Databse
 4. Firebase Storage
- Hardware : Any 64 bit operating System with compatible hardware.

5.2 Risk Management

5.2.1 Risk Identification

1. Scope Risk
2. Schedule Risk
3. Integration Risk
4. Technology Risk

5.2.2 Risk Analysis

1. Scope Risk : The continuous change of requirements in interative manner were considered in design.
2. Schedule Risk : The time for study and implementation were found out and extra 1 months time was kept as backup month.
3. Integration Risk : The modular development design was kept consistent and flexible so that integration of various functionalities is not a problem.
4. Technology Risk : Technology support for a new module and improvements is considered.

5.2.3 Overview of Risk Mitigation, Monitoring, Management

Risk	Category	Probability	Impact	RMMM
Clear understanding of the requirements	Project	50	Critical	Mitigation
Are skilled members available	Technical	50	Critical	Monitoring
Will the technology meet the expectations	Technical	50	Critical	Mitigation
Is the project scope stable	Project	40	Critical	Mitigation
Do all customers know the importance of the product	Business	30	Marginal	Monitoring

Table 5.1: Risk Identification.

5.3 Project Schedule

5.3.1 Project Task Set

1. Literature Survey (T1)
2. Creating Development Environment (T2)
3. Developing Frontend (T3)
4. Creating Storage Environment on Firebase Database (T4)
5. Developing Backend (T5)
6. Integration of Frontend and Backend (T6)
7. Drafting Documentation (T7)

5.3.2 Task Network

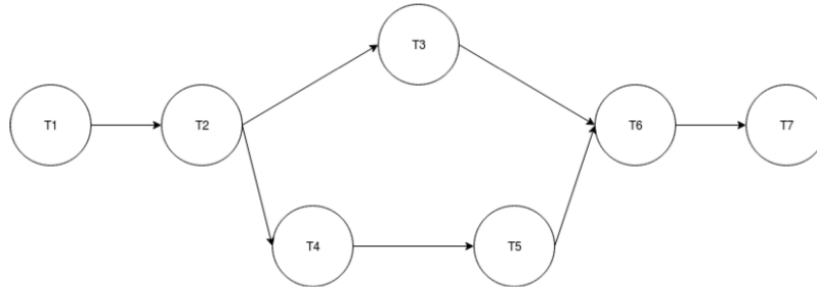


Figure 5.1: Task Network

5.4 Team Organization

The manner in which a team is organized and the mechanisms for reporting are noted. Updates regarding the improvement of project are given to the guide. Also once or twice a month, a meet is held with the project guide regarding updates.

5.4.1 Team Structure

The team structure for the project is identified. Roles are defined.

- Shruti Pawar : Social Distancing module development and Documentation
- Vrushali Ghanekar: Face_recognition model development and Documentation
- Rohan Phatak : Face Mask Module Development and Documentation
- Kashinath Vhanmane : Backend Development and Social Distancing module development.

5.4.2 Management reporting and communication

- Team is always in contact with the project guide.
- Team members also collaborate in-person and work together to improve efficiency. Also, chat groups, mails, and online meetings are used for communication.

CHAPTER 6

PROJECT IMPLEMENTATION

6.1 Overview of Project Modules

6.1.1 Face Mask Module

The Face Mask Module is the very first module of our system which is used to detect whether user has wear a mask properly or not. This module uses SSD algorithm which is an object detection algorithm which will detect mask as an object over the person's face. Two major tasks are performed under face Mask module as follows:

6.1.1.1 Face Mask Detection

In this, actual process of mask detection occurs. Here using SSD object detection algorithm the model will predict whether user has wear mask or not on the given input data.

6.1.1.2 Face recognition

In this, face recognition occurs. Suppose in the earlier process if any user found who has not wear mask properly then, this sub module will recognise his/her face by comparing it with the data present in the database.

6.1.2 Social Distancing Module

The 2nd module in our system is Social Distancing module. In this module, the model detects the violations regarding social distancing. How it works is that from the input frames suppose a frame comes where it consist of group of people having very low distance between them, our model will consider this frame in our system as a violated frame and will save it locally.

6.1.3 Vaccination

Third and last module in our system is the Vaccination module. Task performed by this is to calculate the next probable dose date. What this module will do is that it will take current dose date and will calculate the next dose date and that date will get displayed and even will get sent to user via email.

6.2 Tools and Technologies Used

- Visual Studio
- Pycharm editor

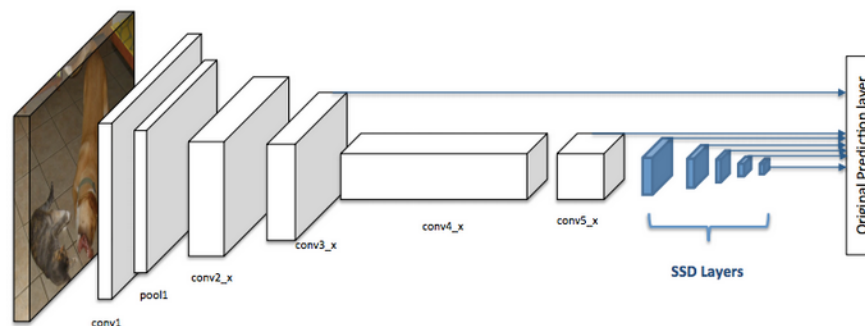
6.3 Algorithm Details

6.3.1 SSD

Working of SSD

SSD has two components: a backbone model and SSD head. Backbone model usually is a pre-trained image classification network as a feature extractor. This is typically a network like ResNet trained on ImageNet from which the final fully connected classification layer has been removed. We are thus left with a deep neural network that is able to extract semantic meaning from the input image while preserving the spatial structure of the image albeit at a lower resolution. For ResNet34, the backbone results in a 256 7x7 feature maps for an input image. We will explain what feature and feature map are later on. The SSD head is just one or more convolutional layers added to this backbone and the outputs are interpreted as the bounding boxes and classes of objects in the spatial location of the final layers activations.

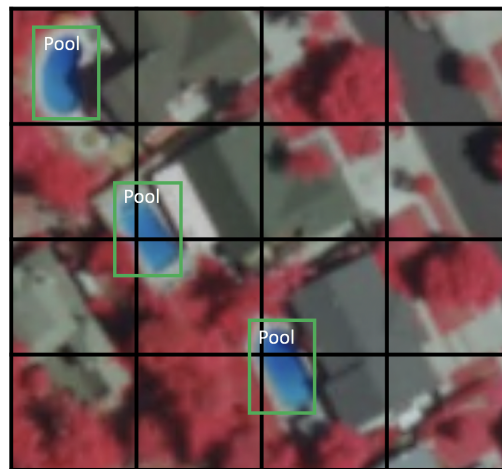
In the figure below, the first few layers (white boxes) are the backbone, the last few layers (blue boxes) represent the SSD head.



Grid cell

Instead of using sliding window, SSD divides the image using a grid and

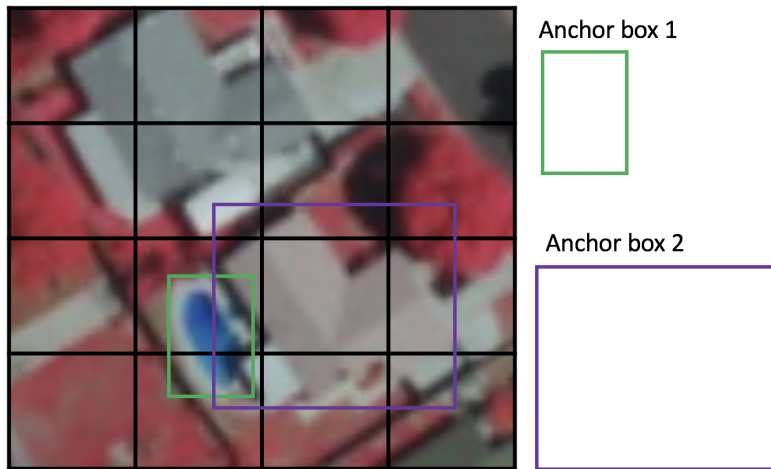
have each grid cell be responsible for detecting objects in that region of the image. Detection objects simply means predicting the class and location of an object within that region. If no object is present, we consider it as the background class and the location is ignored. For instance, we could use a 4x4 grid in the example below. Each grid cell is able to output the position and shape of the object it contains.



Now you might be wondering what if there are multiple objects in one grid cell or we need to detect multiple objects of different shapes. There is where anchor box and receptive field come into play.

Anchor box

Each grid cell in SSD can be assigned with multiple anchor/prior boxes. These anchor boxes are pre-defined and each one is responsible for a size and shape within a grid cell. For example, the swimming pool in the image below corresponds to the taller anchor box while the building corresponds to the wider box.



SSD uses a matching phase while training, to match the appropriate anchor box with the bounding boxes of each ground truth object within an image. Essentially, the anchor box with the highest degree of overlap with an object is responsible for predicting that object's class and its location. This property is used for training the network and for predicting the detected objects and their locations once the network has been trained. In practice, each anchor box is specified by an aspect ratio and a zoom level.

Aspect ratio

Not all objects are square in shape. Some are longer and some are wider, by varying degrees. The SSD architecture allows pre-defined aspect ratios of the anchor boxes to account for this. The ratios parameter can be used to specify the different aspect ratios of the anchor boxes associates with each grid cell at each zoom/scale level.



6.3.2 YOLO V3

YOLO is an abbreviation for the term ‘You Only Look Once’. This is an algorithm that detects and recognizes various objects in a picture (in real-time). Object detection in YOLO is done as a regression problem and provides the class probabilities of the detected images.

YOLO algorithm employs convolutional neural networks (CNN) to detect objects in real-time. As the name suggests, the algorithm requires only a single forward propagation through a neural network to detect objects. This means that prediction in the entire image is done in a single algorithm run.

Importance of YOLO

YOLO algorithm is important because of the following reasons:

Speed: This algorithm improves the speed of detection because it can predict objects in real-time.

High accuracy: YOLO is a predictive technique that provides accurate results with minimal background errors.

Learning capabilities: The algorithm has excellent learning capabilities that enable it to learn the representations of objects and apply them in object detection.

Working of YOLO

YOLO algorithm works using the following three techniques:

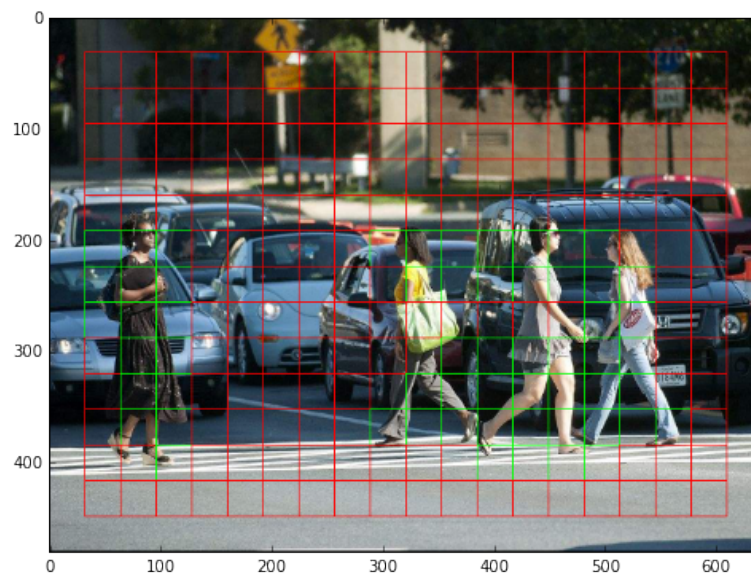
Residual blocks

Bounding box regression

Intersection Over Union (IOU)

Residual blocks

First, the image is divided into various grids. Each grid has a dimension of $S \times S$. The following image shows how an input image is divided into grids.



In the image above, there are many grid cells of equal dimension. Every grid cell will detect objects that appear within them. For example, if an object center appears within a certain grid cell, then this cell will be responsible for detecting it.

Bounding box regression

A bounding box is an outline that highlights an object in an image.

Every bounding box in the image consists of the following attributes:

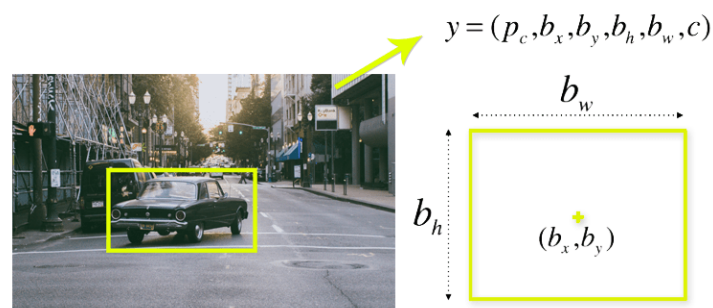
Width (bw)

Height (bh)

Class (for example, person, car, traffic light, etc.)- This is represented by the letter c.

Bounding box center (bx,by)

The following image shows an example of a bounding box. The bounding box has been represented by a yellow outline.

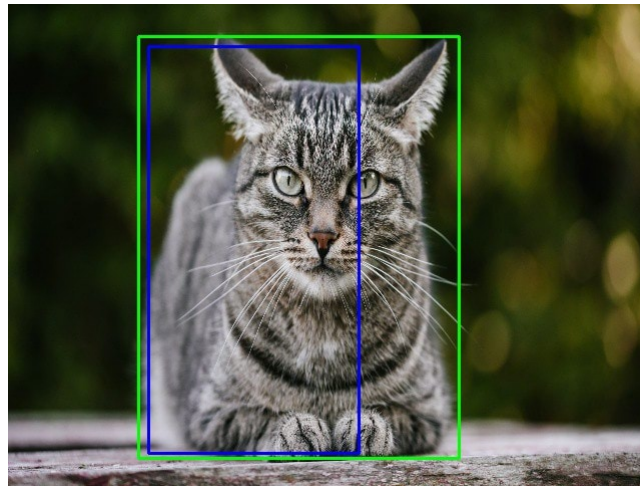


YOLO uses a single bounding box regression to predict the height, width, center, and class of objects. In the image above, represents the probability of an object appearing in the bounding box.

Intersection over union (IOU)

Intersection over union (IOU) is a phenomenon in object detection that describes how boxes overlap. YOLO uses IOU to provide an output box that surrounds the objects perfectly.

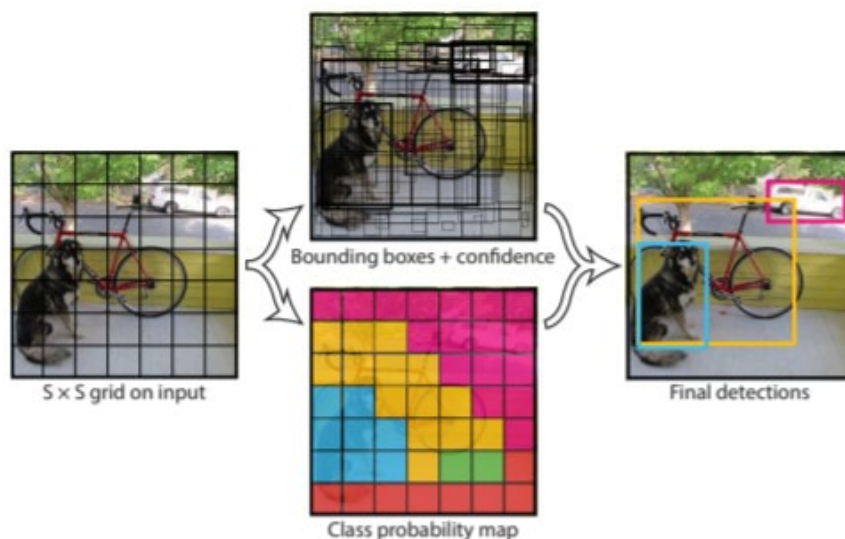
Each grid cell is responsible for predicting the bounding boxes and their confidence scores. The IOU is equal to 1 if the predicted bounding box is the same as the real box. This mechanism eliminates bounding boxes that are not equal to the real box.



In the image above, there are two bounding boxes, one in green and the other one in blue. The blue box is the predicted box while the green box is the real box. YOLO ensures that the two bounding boxes are equal.

Combination of the three techniques

The following image shows how the three techniques are applied to produce the final detection results.



First, the image is divided into grid cells. Each grid cell forecasts B bounding boxes and provides their confidence scores. The cells predict the class

probabilities to establish the class of each object.

For example, we can notice at least three classes of objects: a car, a dog, and a bicycle. All the predictions are made simultaneously using a single convolutional neural network.

Intersection over union ensures that the predicted bounding boxes are equal to the real boxes of the objects. This phenomenon eliminates unnecessary bounding boxes that do not meet the characteristics of the objects (like height and width). The final detection will consist of unique bounding boxes that fit the objects perfectly.

For example, the car is surrounded by the pink bounding box while the bicycle is surrounded by the yellow bounding box. The dog has been highlighted using the blue bounding box.

6.3.3 Dlib Face-recognition

Face_recognition” API is built using dlib’s face recognition algorithms and it allows the user to easily implement face detection, face recognition and even real-time face tracking in your projects or from the command line.

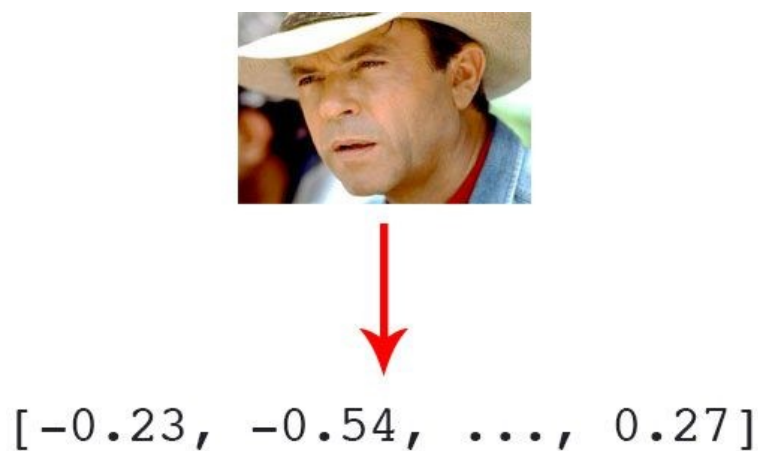
Face Encoding:

The face_recognition API generates face encodings for the face found in the images. A face encoding is basically a way to represent the face using a set of 128 computer-generated measurements. Two different pictures of the same person would have similar encoding and two different people would have totally different encoding. After all the face encodings are generated, Support Vector Classifier (SVC) with scikit-learn is trained on the face encodings along with their labels from all the known faces in the training directory. Finally, the API detects all the faces in the test image you provide and the trained SVC predicts all the known faces in the test image.

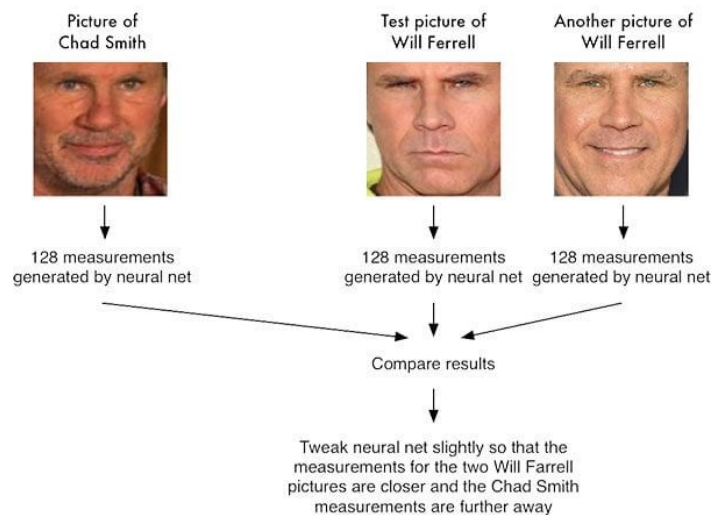
This tool maps an image of a human face to a 128 dimensional vector space where images of the same person are near to each other and images from different people are far apart. Therefore, you can perform face recognition by mapping faces to the 128D space and then checking if their Euclidean

distance is small enough.

When using a distance threshold of 0.6, the dlib model obtains an accuracy of 99.38% on the standard LFW face recognition benchmark, which is comparable to other state-of-the-art methods for face recognition as of February 2017. This accuracy means that, when presented with a pair of face images, the tool will correctly identify if the pair belongs to the same person or is from different people 99.38% of the time.



A single 'triplet' training step:



CHAPTER 7

SOFTWARE TESTING

7.1 Types of Testing

- Unit Testing :

There are multiple units components in our model. Unit testing is a type of testing used in which individual unit or group of related units are tested. So we have tested the individual modules of our system such as the quiz module, Search place module, Group Recommender module and the user profile module. In case user have clicked on some place then that place gets added to recently viewed place or not this we have done in unit testing part.

- Integration Testing :

It is a phase in software testing in which the units or modules are combined and tested as a group. This testing done after completion of unit testing. Here we combine the modules which are providing the same kind of functionality in a single unit and then we performed testing on it like we have combine the user profile with group recommender part so that user can join the different groups. We performed the testing on this module to check whether it is giving the group recommendation to the individual user or not.

- System Testing :

System testing is performed on the entire system, either in accordance with system requirement requirements, functional requirement specifications, or both. System testing examines the system's design and behaviour, as well as the customer's expectations. we have combined the separated modules into a single unit as a system and performed the system testing on it to check whether the system is providing the accurate recommendation to the user in any case if the user is new or existing.

7.2 Test Cases

No.	Functional Test Cases	Actual Output	Expected Output	Test Status
1	Verify if Admin is able to login with appropriate credentials.	Admin logged in successfully.	Admin logged in successfully.	Pass
2	Verify if the Admin is able to see the complete UI of the system.	UI is visible.	UI is visible.	Pass
3	Verify if Admin can click on particular section on UI Window and that respective tab will get opened .	Admin is able to click the tab he/she intends to open and that tab gets opened.	Admin is able to click the tab he/she intends to open and that tab gets opened.	Pass
4	Verify if Admin is able to enter new user details along with users photo.	Admin is able to enter new users data along with his/her 's photo.	Admin is able to enter new users data along with his/her 's photo.	Pass
5	Verify if application is able to show visualization for different types of violations using bar plot and pie plot.	Application successfully displays the violations using mentioned plots.	Application successfully displays the violations using mentioned plots.	Pass
6	Verify if email notification/alert gets sent to the user identified by system.	Email Notification/alert gets sent successfully.	Email Notification/alert gets sent successfully.	Pass
7	Verify if the profile page is working fine	Display profile details correctly.	Display profile details correctly.	Pass

Table 7.1: Unit Testing Test Cases

No.	Functional Test Cases	Actual Output	Expected Output	Test Status
1	To check whether the application chooses the selected module and loads pages accordingly.	The pages from ADMIN Module are loaded.	The pages from ADMIN Module are loaded.	Pass
2	To check whether the application stores data according to security constraints and displays the violation data with respect to two modules in system (i.e. Face Mask and Social Distancing Module) along with date of violation.	Successfully displays List of violations according to the criteria specified.	Successfully displays List of violations according to the criteria specified.	Pass

Table 7.2: Integration Testing Test Cases

No.	Functional Test Cases	Actual Output	Expected Output	Test Status
1	To check whether the application runs on the local machine	Should Display the Integrated Development Environment	Application is running	Pass
2	Verify if while processing or when the software is in running state UI window is still accessible.	User can still access UI window while system is in running state.	User can still access UI window while system is in running state.	Pass

Table 7.3: System Testing Test Cases

CHAPTER 8

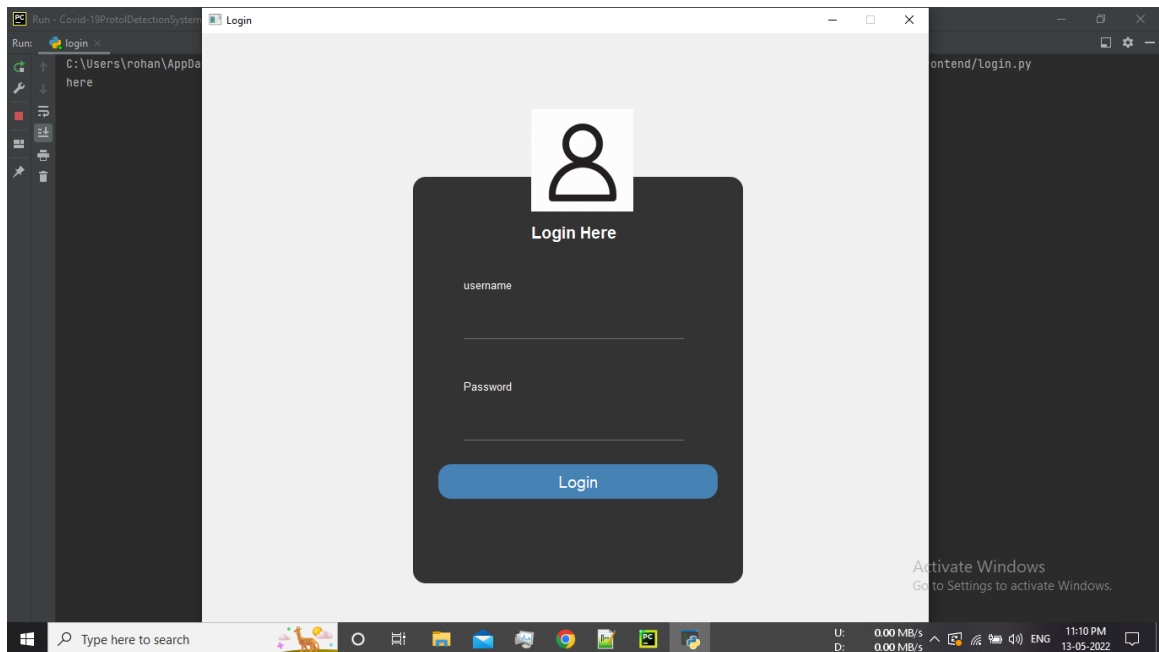
RESULTS

8.1 Outcomes

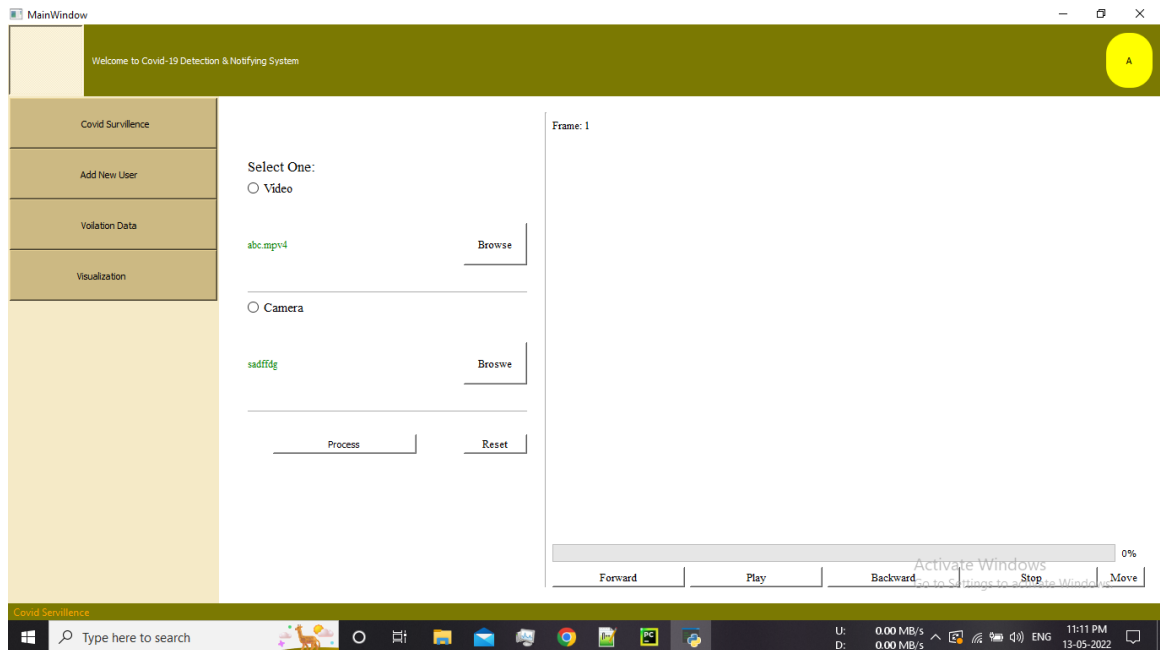
- Admin can provide the video as input to our system and can see the violations happened in that video. Such violated frames will get stored locally temporarily.
- Admin can add new user/employee's details into the system
- The Admin can see the violated frames for each type of violation along with their respective visualization.
- Database for maintaining records of users/employees, their data, violated frames.

8.2 Screen Shots

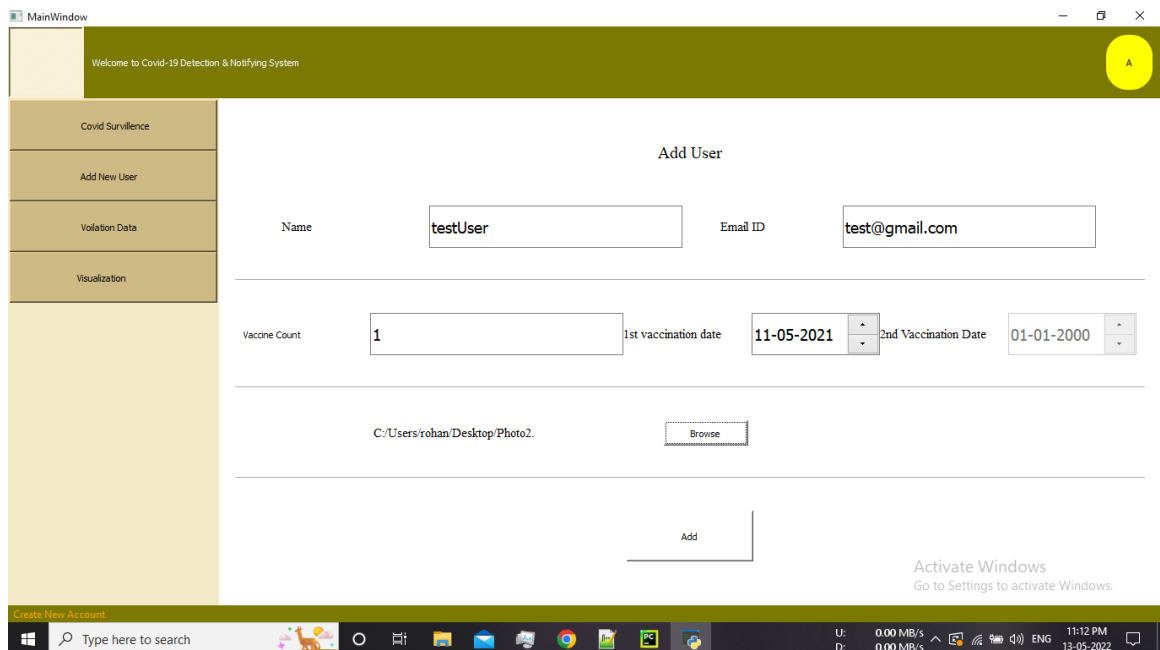
8.2.1 Admin Login:

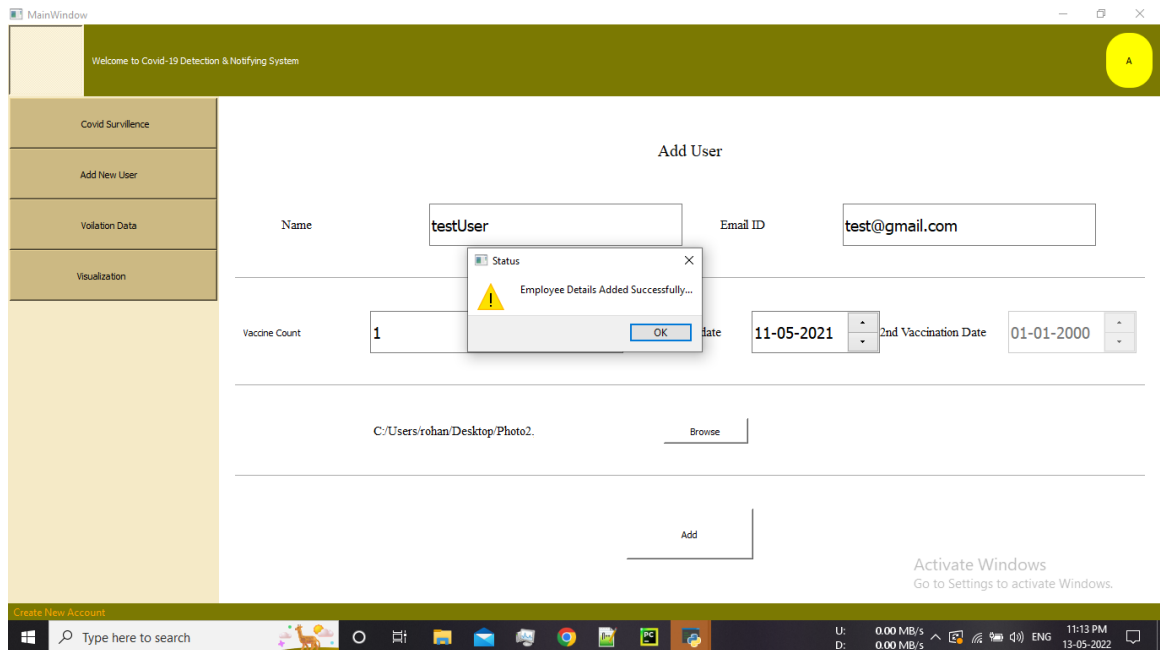


8.2.2 Main System UI

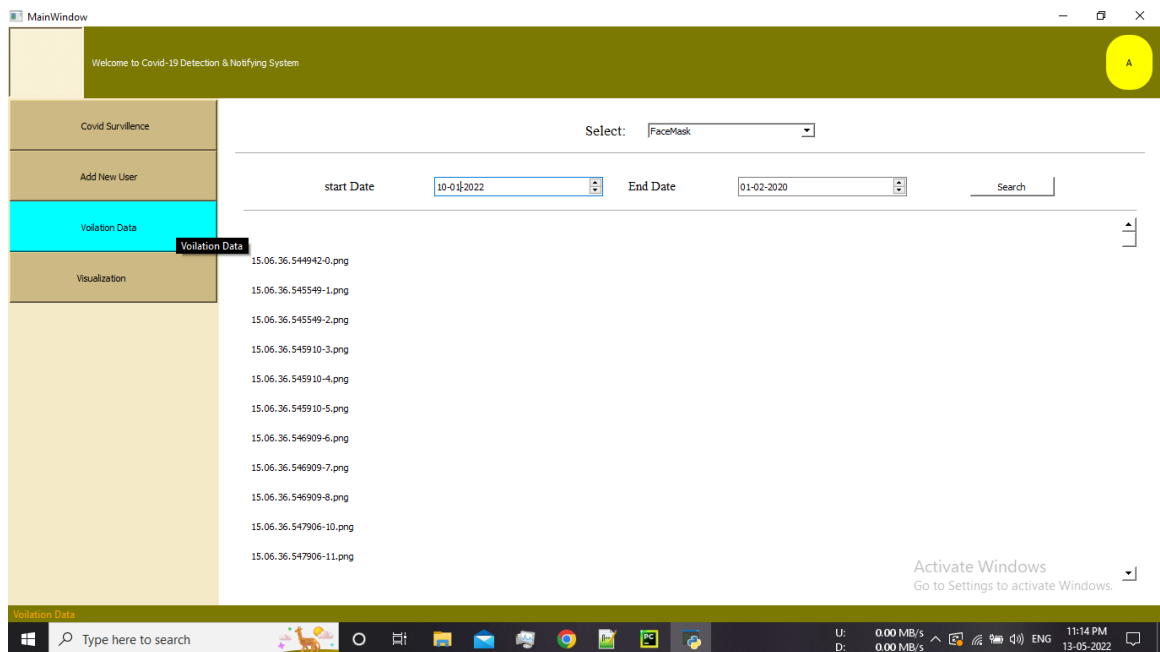


8.2.3 Add User Section

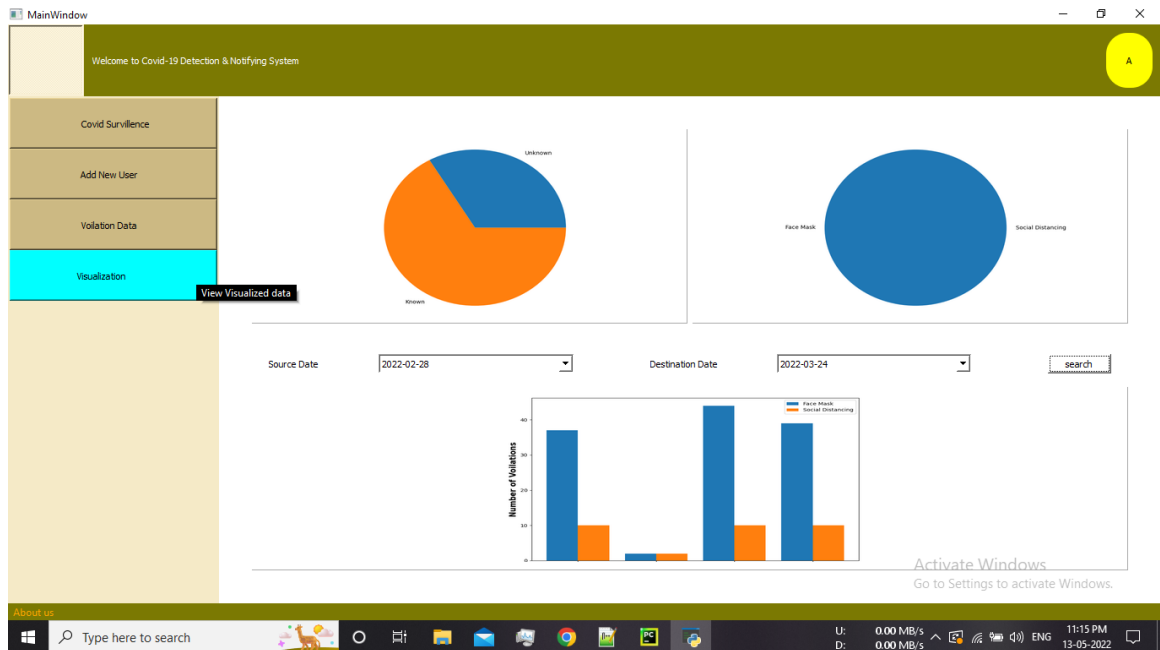




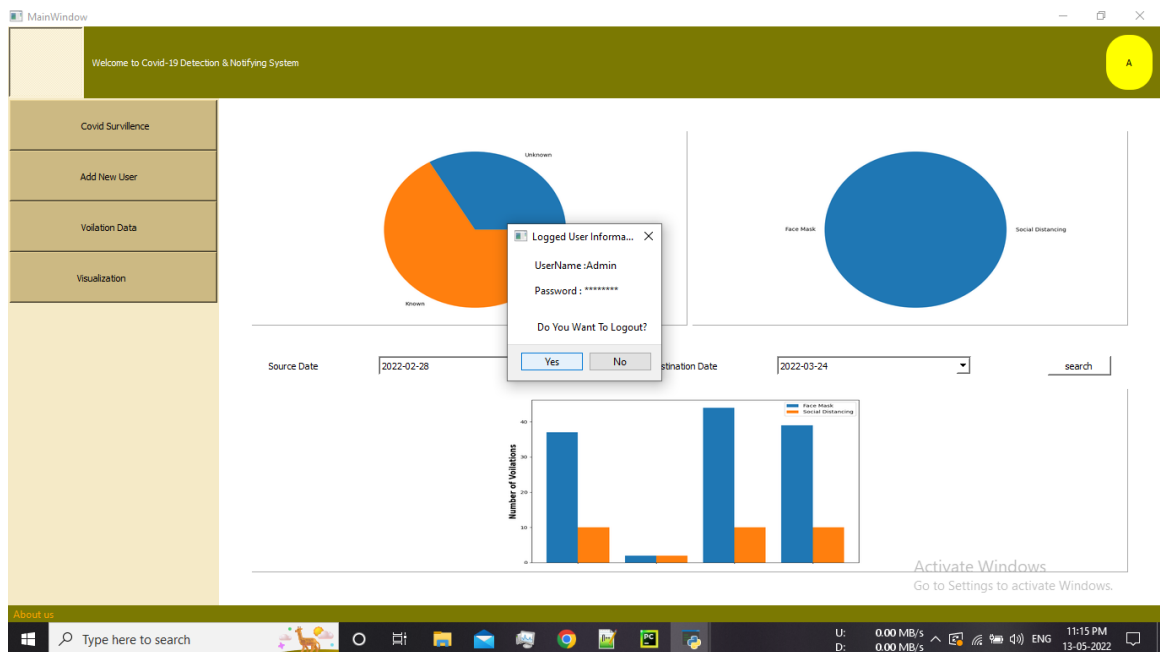
8.2.4 Violation Data Section



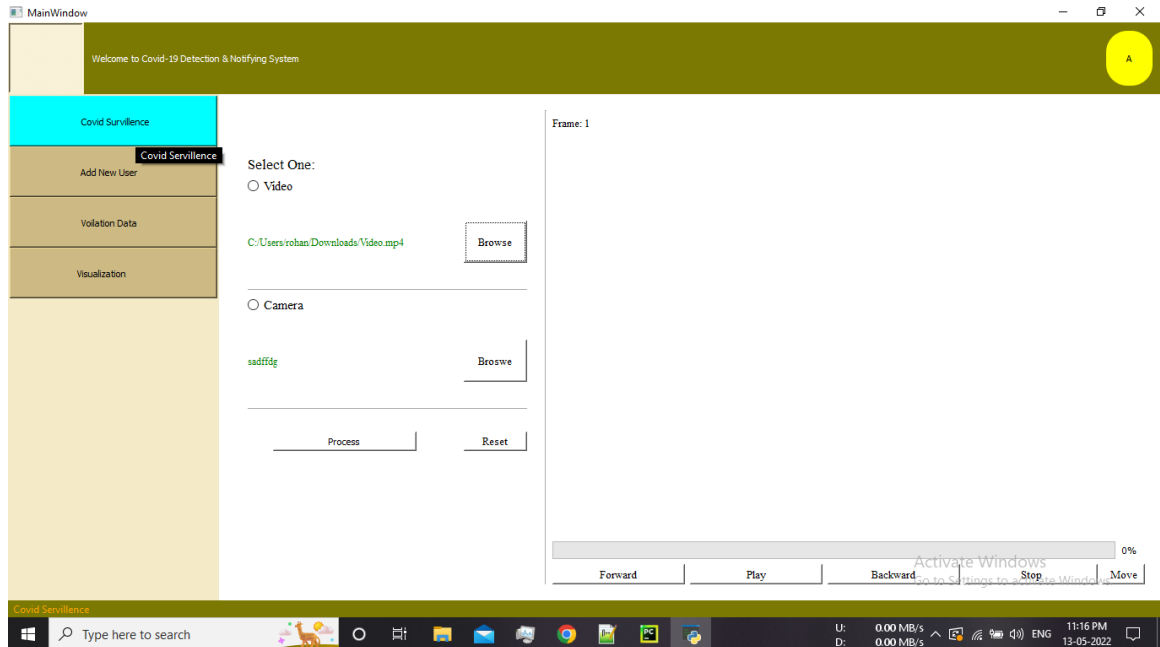
8.2.5 Visualization Section



8.2.6 Profile Section

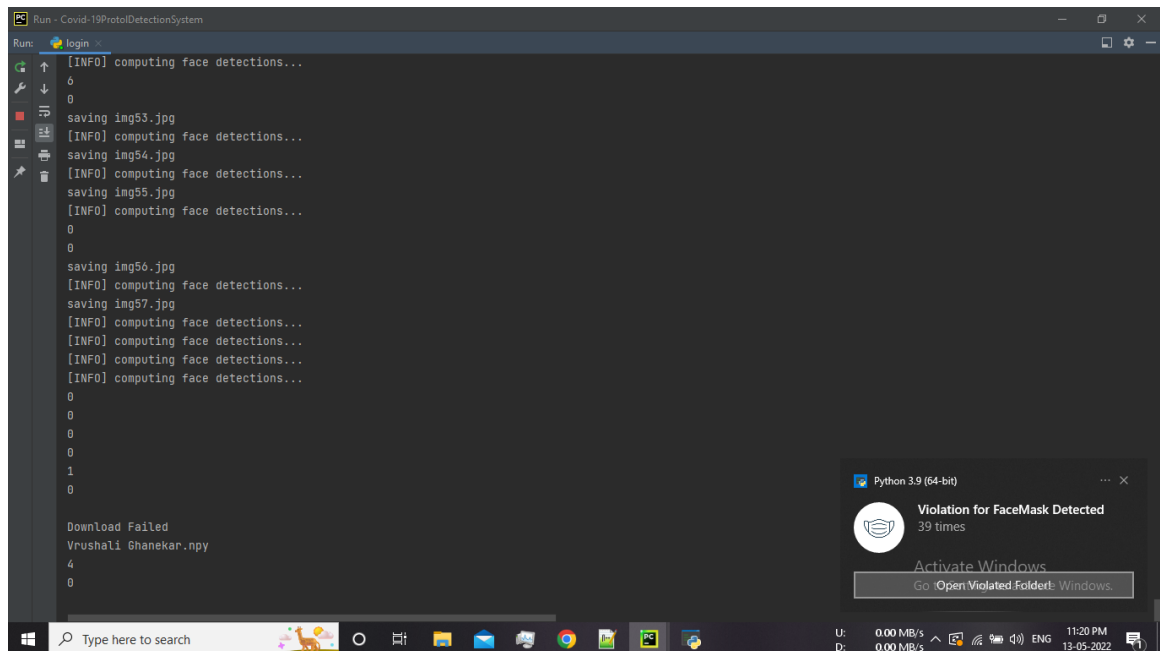


8.2.7 Covid Surveillance Section

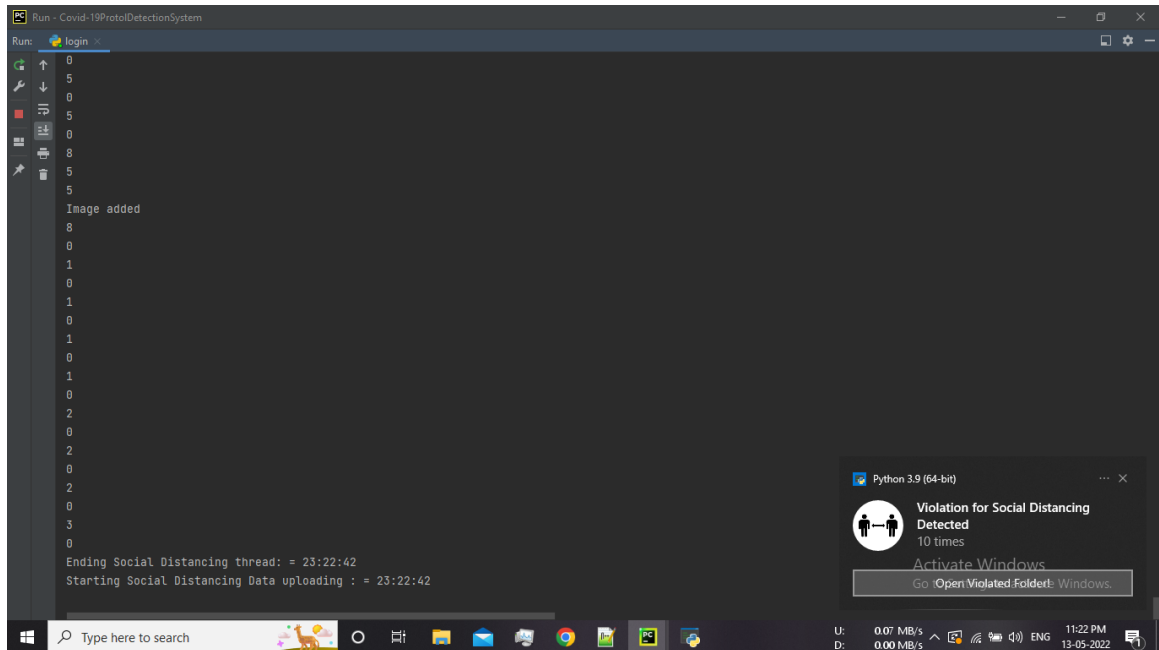


8.2.8 Desktop Notification

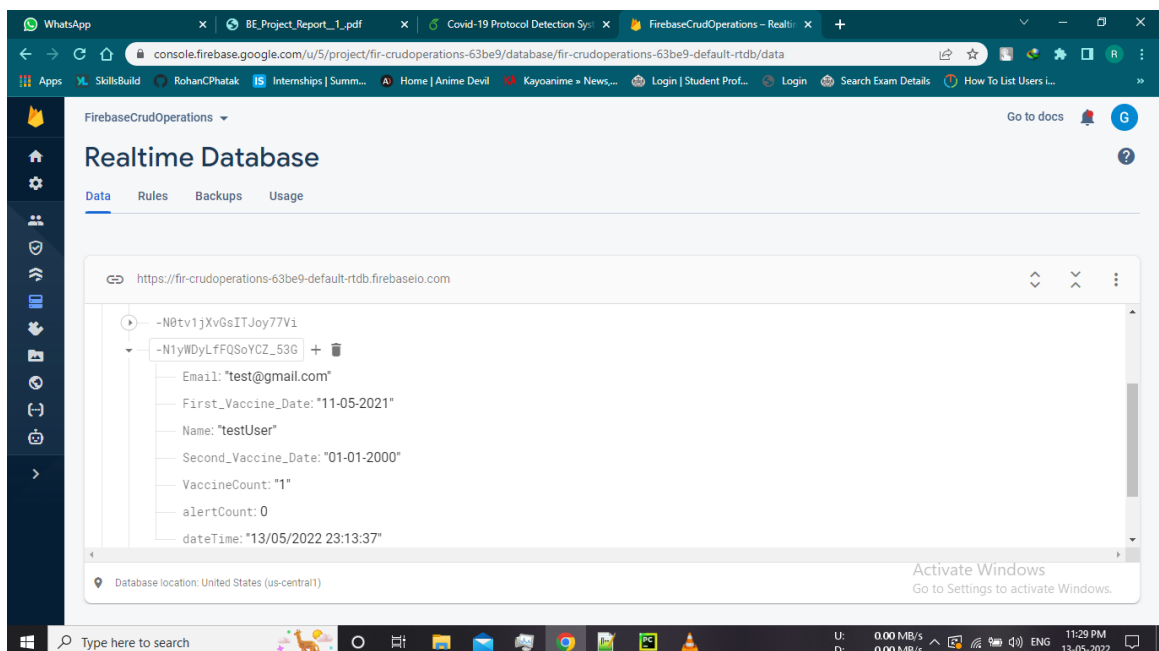
8.2.8.1 Face Mask Detection



8.2.8.2 Social Distancing Violation

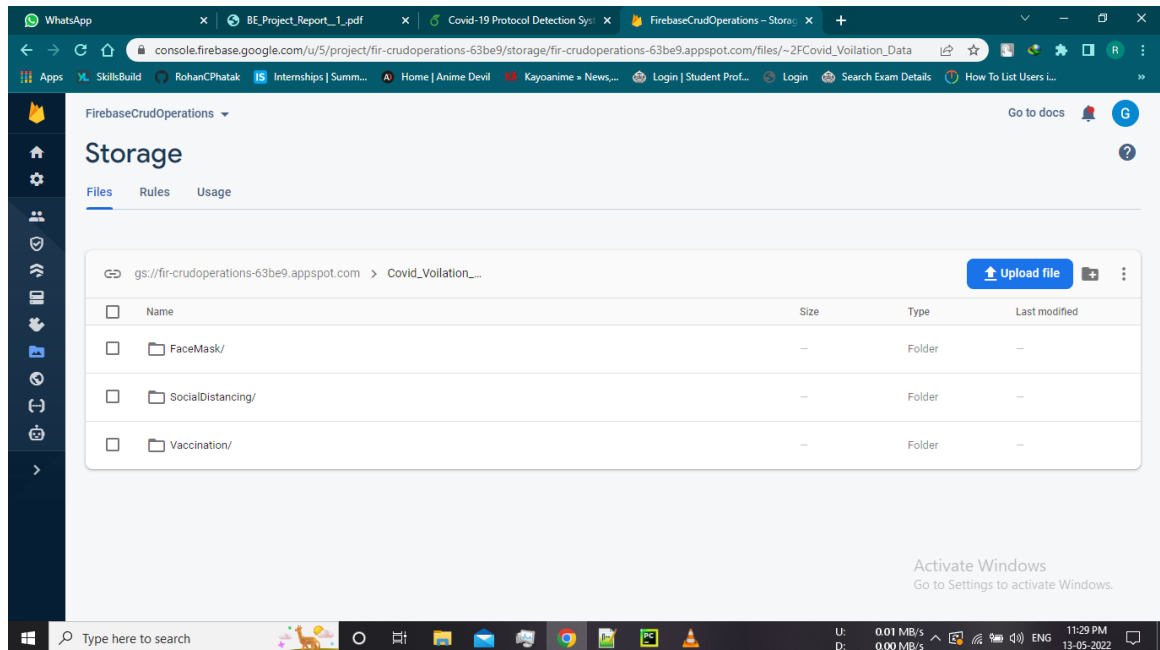


8.2.9 Database Structure

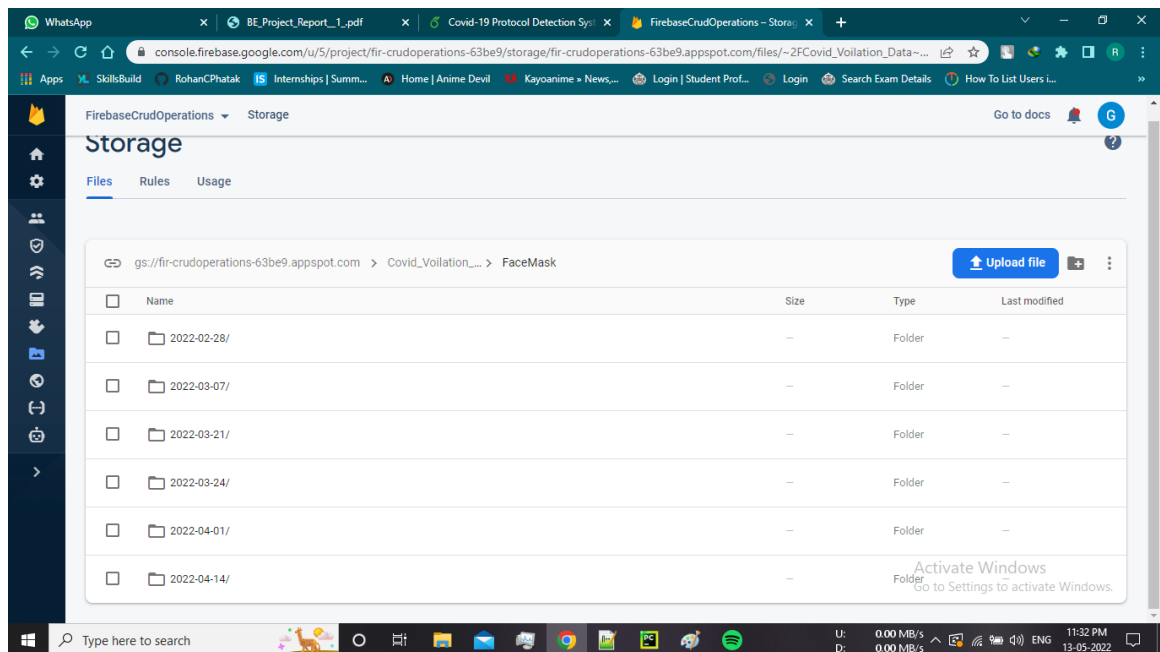


8.2.10 Violated Data Storage

8.2.10.1 Violation type wise divication



8.2.10.2 Date Wise storage of violated frames



CHAPTER 9

CONCLUSION

9.1 Conclusion

The proposed and implemented system aims to help people in private sectors to follow the rules. We have provided the best possible solution to reduce the problem of violation of Covid-19 rules. The major concern that private sector facing is, proper follow of covid-19 protocols. Through this system, we can monitor the people in the organization that leads to help follow the rules and also reduce the new covid-19 cases. System also have functionality of alert, so that people gets the notification if they unfollow the rules.

9.2 Future Work

1. Updating used model:- The currently used models in the system require or work on specific image image quality. The updated models can be used to work on images having very high or very low quality.
2. Implementing prediction capability for social distancing:- Currently the system is only capable for detecting violation of social distancing rules but in future we can expand this system to make it capable for predicting the social distancing violation based on the live data.
3. Creating lite version:- The currently developed application works well on high configuration hardware but when it comes to run on low configuration hardware it takes comparatively more load to generate the final result. As a future work we can try to make application light weight so that it can work on low configuration devices also.

9.3 Applications

1. Hospitals and Clinics:-
This is an important domain that needs special attention during a pandemic. To safeguard the health of doctors and nurses, this system can be implemented to detect whether a patient is wearing a mask. A CCTV camera can be placed to detect mask and social distancing violations which can in turn help reduce crowding while patients are waiting in queues for their appointments.
2. Airports:-
Many of the destinations around the world have started easing travel restrictions. This system can be implemented in airports where people

are waiting in queues at check-in facilities, security clearance gates, passport control, waiting lobbies etc.

3. Companies:-

As many companies are opening after lockdown this system can be implemented to ensure that employees are following all the COVID 19 safety protocols.

CHAPTER 10

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ANNEXURE A

A problem is NP-hard if an algorithm for solving it can be translated into one for solving any NP-problem (nondeterministic polynomial time) problem. NP-hard therefore means "at least as hard as any NP-problem," although it might, in fact, be harder. Thus the problem statement implies that it is a NP-hard problem.

ANNEXURE B

Plagiarism of project report

